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Method of operating a voice coil motor and corresponding control circuit

Abstract

Embodiments provide a method of operating a voice coil motor via a transconductance loop. The method includes detecting an actual value of a supply voltage of the transconductance loop. An offset compensation signal of the transconductance loop is produced as a function of the detected actual value of the supply voltage based on a relationship between offset values and the supply voltage of the transconductance loop. The offset compensation signal is applied to a loop control signal of the transconductance loop. A drive current is applied to the voice coil motor. The drive current is related to a target drive current that is based on the loop control signal.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a translation of and claims the priority benefit of Italian patent application number 102023000001830, filed on Feb. 6, 2023, which is hereby incorporated by reference to the maximum extent allowable by law.

TECHNICAL FIELD

(2) The description relates to hard disk drives (HDDs). One or more embodiments can be applied, for instance, to hard disk drives used in processing devices such as computers, servers, data centers and the like.

BACKGROUND

(3) Hard disk drives (HDDs) employ a spindle motor to rotate one or more disks as well as voice coil motors (VCMs) to move the heads with respect to the disk(s). That is, the spindle motor is used to spin the hard disk(s) on which data are stored and the VCM motor is used to position the read/write heads on the surface of hard disk(s). Hard disk drives thus include at least two motors (spindle and VCM) that are driven by power semiconductor devices (integrated circuits) usually referred to as power combos.

(4) Positioning of the heads at the surface of the disks is desired to take place with high precision. This result is facilitated by a micro-controller that controls a position loop. In order to control the position of the heads, the controller modulates the current in the VCM motor through a transconductance amplifier that converts an input voltage (e.g., from a digital-to-analog converter DAC) into an output current.

(5) Offset calibration of the transconductance loop facilitates an accurate control of the load current. Offset calibration is performed once, before the VCM usage. An issue with calibration lies in that the result is a function of supply voltage value, in so far as offset may vary during normal VCM operations if the supply voltage is not constant. Accurate calibration of the offset of such a transconductance loop as a function of variations in the supply voltage is thus a desirable feature of a hard disk drive.

SUMMARY

- (6) Embodiments can contribute in facilitating such accurate calibration.
 - (7) One or more embodiments relate to a corresponding controller circuit.
 - (8) One or more embodiments relate to a corresponding hard disk drive.
 - (9) One or more embodiments relate to a corresponding processing device. A computer, a server, or data center equipped with a hard disk drive are exemplary of such a device.
 - (10) One or more embodiments provide for dynamic compensation of the offset of a transconductance amplifier based on the measurement of the supply voltage.
 - (11) Certain embodiments involve determining at least two offset values for two different supply voltage values, e.g., two different voltage values obtained from the supply voltage applied as a common-mode voltages at the inputs of the sense amplifier.
 - (12) In certain embodiments, an instantaneous value for the offset is stored as obtained via a look-up table or as calculated based on a function that links the common mode voltage to a related offset.
 - (13) In certain embodiments, two offset measurements are performed and the function is the equation of a straight line passing through two points.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) One or more embodiments will now be described, by way of example only, with reference to the annexed figures, wherein:
- (2) FIG. 1 is a block diagram of a current control loop circuitry (“transconductance amplifier”) as used to control a VCM current in a hard disk application,
- (3) FIGS. 2 and 3 are diagrams exemplary of ideal and actual behavior of VCM current control loops,
- (4) FIG. 4 is exemplary of conventional offset calibration and compensation in a transconductance amplifier in current control loop circuitry as illustrated in FIG. 1,
- (5) FIGS. 5A and 5B are illustrative of offset measurement according to embodiments of the present description,
- (6) FIG. 6 is a block diagram exemplary of how results of offset measurement as exemplified FIGS. 5A and 5B can be used for offset compensation according to embodiments of the present description;
- (7) FIG. 7 is an exemplary diagram of loop offset compensation in embodiments of the present description;
- (8) FIG. 8 is a flow-chart of a loop offset calculation and compensation procedures according to embodiments of the present description; and
- (9) FIGS. 9A and 9B are combined diagrams comparative of conventional offset compensation (FIG. 9A) and offset compensation according to embodiments of the present description (FIG. 9B).
- (10) Corresponding numerals and symbols in the different figures generally refer to corresponding parts unless otherwise indicated.
- (11) The figures are drawn to clearly illustrate the relevant aspects of the embodiments and are not necessarily drawn to scale. The edges of features drawn in the figures do not necessarily indicate the termination of the extent of the feature. Also, for the sake of simplicity and ease of explanation, a same designation may be applied throughout this description to designate a circuit node or line as well as a signal occurring at that node or line.
- (12) Any quantitative values indicated in the figures are examples and non-limiting.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

- (13) In the ensuing description, various specific details are illustrated in order to provide an in-depth understanding of various examples of embodiments according to the description. The

embodiments may be obtained without one or more of the specific details, or with other methods, components, materials, etc. In other cases, known structures, materials, or operations are not illustrated or described in detail so that various aspects of the embodiments will not be obscured. (14) Reference to “an embodiment” or “one embodiment” in the framework of the present description is intended to indicate that a particular configuration, structure, or characteristic described in relation to the embodiment is comprised in at least one embodiment. Hence, phrases such as “in an embodiment,” “in one embodiment,” or the like, that may be present in various points of the present description do not necessarily refer exactly to one and the same embodiment. Furthermore, particular configurations, structures, or characteristics may be combined in any adequate way in one or more embodiments.

(15) The headings/references used herein are provided merely for convenience and hence do not define the extent of protection or the scope of the embodiments.

(16) As discussed in the introductory portion of this description, HDD is an acronym for Hard Disk Drive.

(17) A hard disk drive is a component of various types of processing devices such as personal computers, servers, data centers or the like and is the physical location where information is stored.

(18) Hard disk drives (HDDs) employ a spindle motor to rotate one or more disks as well as voice coil motors or VCMs to move the heads with respect to the disk(s).

(19) FIG. 1 is a block diagram of current control loop circuitry (as embedded in a power “combo”) used in a hard disk application.

(20) More specifically, FIG. 1 shows an example of a conventional transconductance amplifier used to drive a VCM motor in a hard disk drive HDD by driving a current $I_{\text{sub.vcm}}$ through the motor windings.

(21) The arrangement of FIG. 1 essentially comprises a power amplifier **10** that drives the VCM motor via two lines VCMN and VCMP with an amperometry sensor (such as a sensing resistor R_s , arranged between the motor and one of the lines VCMN and VCMP) configured to produce a signal indicative of the intensity of the current $I_{\text{sub.vcm}}$ through the motor.

(22) As illustrated in FIG. 1, the power amplifier **10** comprises two power stages **101** (with an output line VCMN) and **102** (with an output line VCMP) each comprising a differential amplifier such as an operational amplifier or op-amp supplied via a supply voltage V_M , e.g., referred to ground.

(23) Each of the power stages **101** and **102** has a first input (e.g., inverting for the stage **101** and non-inverting for the stage **102**) configured to be coupled (e.g., via a voltage divider, illustrated and not expressly labelled for simplicity) to a reference node at a voltage V_{ref} .

(24) The supply voltage V_M and the reference voltage V_{ref} are generated in a manner known per se to those of skill in the art.

(25) Each of the power stages **101** and **102** have second inputs (e.g., non-inverting for the stage **101** and inverting for the stage **102**) configured to be coupled (e.g., via a voltage divider/coupling resistor, again illustrated and not expressly labelled for simplicity) to a common node A at the output of an error amplifier **12** (again an operational amplifier or op-amp, for instance).

(26) The error amplifier **12** has a first input (e.g., non-inverting) configured to be coupled (e.g., via a coupling resistor, illustrated and not expressly labelled for simplicity) to a reference node at a reference voltage (this may be assumed to be again the reference voltage V_{ref} , for simplicity).

(27) The error amplifier **12** has a second input (e.g., inverting) coupled: i) via a line including a resistor R_c and a capacitor C_c , to the output node A; ii) via an input resistor R_{in} , to the output of a digital-to-analog converter DAC **14** (receiving the voltage V_{ref}) providing a voltage $V_{\text{sub.DAC}}$; and iii) via a coupling resistor R_f to the output of a sense amplifier **16**.

(28) The sense amplifier **16** (once more, an operational amplifier or op-amp, for instance) has a first and second inputs (e.g., non-inverting and inverting) coupled (e.g., via a coupling resistors R) to the opposed ends of the sensing resistor R_s . Across the resistor R_s a voltage ($R_s \cdot I_{\text{sub.VCM}}$) can

be sensed which is indicative of the intensity of the current $I_{\text{sub.vcm}}$ across the motor VCM.

(29) Resistors KR (assumed to be of equal value for simplicity) are provided between the output and the inverting input of the sense amplifier **16** and between the non-inverting input of the sense amplifier **16** and a reference node at a reference voltage (this may be assumed to be again the reference voltage V_{ref} , for simplicity).

(30) As illustrated, the sense amplifier output voltage is referred to V_{ref} , this means the sense amplifier output voltage is equal to V_{ref} if $I_{\text{sub.VCM}}=0$ and is modulated across V_{ref} if $I_{\text{sub.VCM}}$ is not zero.

(31) FIG. **1** is thus exemplary of a VCM driver, which is configured as a transconductance amplifier comprised of an (e.g., DMOS) H-bridge power stage **101**, **102**, an error amplifier **12**, a current sense **16** amplifier, and an (e.g., 15-bit linear) voltage DAC converter **14**.

(32) As illustrated, the DAC converter **14** output voltage is referred to V_{ref} . The DAC output voltage is thus equal to V_{ref} if the digital word input to the DAC input is zero and is modulated across V_{ref} in response to the DAC input word “DW” being different from zero. In the exemplary case considered here, the power amplifier (indicated as 10 as whole) can be a class AB voltage amplifier.

(33) The error amplifier **12** is configured to close the transconductance loop, using feedback from the current sense amplifier **16** and an input command from the DAC converter **14**.

(34) The diagram of FIG. **1** is thus illustrative of a transconductance amplifier that includes a circuit configured to measure the current $I_{\text{sub.vcm}}$ in a load (the motor VCM) and includes a sense amplifier **16** (assumed to provide gain G_s) that receives as an input the voltage $R_s \cdot I_{\text{sub.VCM}}$ obtained from the sensing resistor R_s arranged in series to the load (VCM).

(35) The current $I_{\text{sub.vcm}}$ in the load, controlled by the transconductance loop of FIG. **1**, is a function of constant parameters (namely, the values for R_s , G_s , R_f , R_{in}) and of the output value $V_{\text{sub.DAC}}$ of the digital-to-analog converter **14**.

(36) The digital-to-analog converter **14** is configured to receive an input digital word DW (e.g., [14:0]) generated in a manner known per se to those of skill in the art.

(37) This relationship can be expressed in the form:

(38)
$$I_{\text{VCM}} = (R_f / R_{\text{in}}) * [1 / (R_s * G_s)] * V_{\text{DAC}} = \text{DC}_{\text{Gain}} * V_{\text{DAC}}$$

(39) In HDD applications, especially during head tracking operation, a servo controller MCU adjusts the VCM DAC register (DW) so as to drive the correct current to keep the heads on track.

(40) The (e.g., 15-bit) voltage DAC **14** commands the current in the VCM circuit, with the DAC **14** controlled via the controller MCU through a serial port receiving a DAC word DW, [14:0], for instance.

(41) The DAC output voltage is centered on the voltage V_{ref} and is injected into the transconductance loop via the resistor R_{in} . For instance, positive DAC commands can drive positive VCM currents and negative DAC commands negative VCM currents; in certain embodiments, negative DAC commands can drive positive VCM currents and positive DAC commands negative VCM currents.

(42) Positioning of the heads at the surface of the disks of the hard disk drive HDD (which is desired to take place with high precision) is facilitated by the micro-controller MCU associated with the hard disk drive HDD in order to control a position loop: in order to control the position of the heads, the controller MCU modulates the current in the VCM motor through a transconductance amplifier that converts an input voltage (e.g., from a digital-to-analog converter DAC) into an output current.

(43) It will be appreciated that the representation in FIG. **1** (the same also applies to the representations in FIGS. **4**, **5A**, **5B** and **6**) is essentially functional, in that the motor VCM and the parts/elements associated therewith can in fact be incorporated in the hard disk drive HDD, while the associated controller MCU can be either incorporated in the hard disk drive HDD or be a distinct element.

(44) The diagram in FIG. 2 shows an (ideal) transconductance characteristic that gives the VCM current $I_{sub.VCM}$ as a function of the word DW written in the DAC 14.

(45) As noted, this is an ideal characteristic, assuming that a zero DAC value translates into a corresponding zero current driven into the motor (VCM).

(46) Even though these factors do not appear in the formula reported above, the current $I_{sub.vcm}$ in the load is in fact affected by offsets generated by the sense amplifier 16 and by the error amplifier 14.

(47) The actual loop behavior may thus be (quite) different: due to offset in the components of the VCM current loop (primarily the error and sensor amplifiers 12 and 16), a zero DAC value (DAC input word DW) does not result in a corresponding zero VCM current.

(48) This is exemplified in the diagram of FIG. 3, where the “ideal” $I_{sub.VCM}$ v. DAC Word behavior of FIG. 2 is portrayed (in continuous line) against possible actual behaviors for positive offset (chain line) and negative offset (dashed line).

(49) In order to remedy situations as portrayed in FIG. 3, a DAC register value can be used given by the sum (with sign) of the value coming from the servo loop and a value obtained via a zero VCM current offset calibration procedure. In that way, the servo loop is supplied with a “calibrated” correct value that is written into the DAC 14.

(50) A power combo can thus be configured to perform a calibration routine that measures (and eventually cancels) the VCM current control loop offset. Calibrating the offset of the transconductance amplifier may thus facilitate achieving a notionally zero (or very small) offset. That calibration routine can be performed (under the control of the controller MCU, for instance) before track following operation of the heads is started.

(51) In that way an accurate control of the position of the motor (VCM) is facilitated by the transconductance amplifier having an offset that is notionally zero or, in any case, very small.

(52) FIG. 4 (where parts or elements like parts or elements already discussed in connection with FIG. 1 are indicated with like reference symbols) is exemplary of a conventional approach to calibration wherein a zero current is forced in the load (motor VCM) and the inputs of the sense amplifier 16 are forced to a voltage $V_m/2$ that is selected equal to half the supply voltage V_M of the power stage (a different notation is deliberately retained for clarity).

(53) More specifically, FIG. 4 is exemplary of a circuit configuration that can be used for measuring offset in current control loop circuitry, with the results of such offset measurement eventually used for offset compensation.

(54) In (conventional) offset measurement as illustrated in FIG. 4, one branch of the power stage (e.g., the stage 101) is forced to high impedance—in a manner known per se—to have zero current in the load while the other branch of the power stage (e.g., the stage 102) is kept active and forced to a value $V_m/2$ where V_m is equal to the (normal) supply voltage value V_M .

(55) This type of operation can be facilitated via switches Sw1, Sw2 and Sw3 that can be implemented in a manner known per se to those of skill in the art, e.g., as MOSFET transistors and can be operated (turned on=conductive and off=non-conductive) under the control of the controller MCU, for instance.

(56) The switch Sw1 is included in the feedback loop of the error amplifier 12. With switch Sw1 open (that is, non-conductive), the error amplifier 12 behaves as a comparator.

(57) The switch Sw2 is arranged intermediate the node A and the stage 102 and, when open (non-conductive) decouples the stage 102 from the node A, thus from the error amplifier 12. It is noted that the other stage 101 is forced to high impedance to have zero current in the load.

(58) The switch Sw3 is arranged (via coupling resistors that are illustrated and not expressly referenced) across the inputs of the stage 102 and, when closed (conductive), the switch Sw3 causes both these inputs (and not just the non-inverting input) to be coupled to the reference voltage V_{ref} forcing, de facto, a null differential voltage across the inputs of the stage 102. In these conditions the output of the stage 102 is forced to $V_m/2$.

(59) The sense amplifier **16** and the error amplifier **12** are enabled, the output VCMN is tri-stated (at a high impedance HiZ) and the output VCMP is forced to $V_m/2$ (Sw2 is open and Sw3 is closed).

(60) The switch Sw1 is open, allowing the error amplifier to behave as a comparator (in that way, only two voltage levels are allowed, positive or negative saturations).

(61) The output of the error amplifier **12** on a line Error OUT is routed to a serial port register in order to make it readable during calibration (error sign).

(62) With this configuration the VCM DAC **14** can be swept (sweeping the input word DW), and the output of the error amp **12** (error sign) can be monitored via firmware. The loop offset is measured (e.g., as a value of the digital word DW input to the DAC converter **14**) when the error amp output on the line Error OUT (error sign) switches states.

(63) It is noted that with the power stage **10** configured with the VCMN line in tristate (HiZ) and the VCMP line driven at $V_m/2$, the sense resistor R_s is driven at $V_m/2$ too, setting the voltage of the inputs of the sense amplifier **16** at a known voltage amplitude.

(64) This offset calibration (offset measurement) method can be implemented under the control of the controller MCU and is advantageous in HDD applications because the common mode voltage of the sense amplifier **16** inputs equal to $V_m/2$ is equivalent to the common mode voltage of the sense amplifier **16** inputs during the track following operation (track following is performed at low current and both VCM driver outputs VCMP and VCMN are both very close to $V_m/2$).

(65) That is, under the conditions exemplified in FIG. 4: the differential voltage across the sense amplifier **16** is equal to zero (due to the current $I_{sub.VCM}$ being zero) while a common-mode input voltage is applied to the sense amplifier **16** equal to $V_m/2$, and the output of the sense amplifier **16** will be equal to $V_{ref} + \text{offset}$ where the offset indicates a voltage offset value (ideally zero, but in fact different from zero due to construction inaccuracies of the sense amplifier itself).

(66) During calibration of the offset of the transconductance amplifier (offset measurement), the gain of the error amplifier **12** is (markedly) increased by opening the switch Sw1 in the feedback path of the error amplifier **12** as shown in FIG. 6.

(67) This increased gain facilitates the detection of changes of sign of the error which is carried out by analyzing the output of the error amplifier available over a line Error OUT.

(68) Under ideal conditions (zero offset) the signal on the line Error OUT changes polarity when the DAC input word DW moves around zero and, as consequence, the output signal from the DAC converter **14** moves around the reference value V_{ref} (as illustrated the DAC converter **14** is configured to have an output voltage equal to V_{ref} when its input word DW is zero).

(69) Conversely, in real conditions the change in error polarity takes place instead for values of the input word DW other than zero (for DAC output other than V_{ref}), which depend on the offset of the sense amplifier **16** and the offset of the error amplifier **12**: see the diagram of FIG. 3 for reference.

(70) A calibration procedure as exemplified in FIG. 4, therefore, involves a search for a value for the input word DW that, if applied to the DAC converter **14**, produces a change of sign at the output of the error amplifier (line Error OUT).

(71) This value can then be used as an offset compensation signal OC of the transconductance loop. Such an offset compensation signal OC can be applied (e.g., at a summation node **18**—with sign—at the input of the DAC converter **14**) to the loop control signal DW of the transconductance loop **10, 12, 16**.

(72) In FIG. 4 the offset compensation signal OC is represented in dashed lines to highlight the fact that offset compensation is distinct from the offset measurement phase discussed previously.

(73) In fact, offset compensation (injection of the signal OC into the input word DW at the node **18**) takes place with the transconductance loop **10, 12, 16** brought to its normal operating condition, with the switches Sw1 and Sw2 closed—that is, conductive—and the switch Sw3 open—that is, non-conductive—as indicated by dashed lines in FIG. 4.

(74) Operation as discussed so far in connection with FIGS. 1 through 4 is per se conventional in

the art, which makes it unnecessary to provide a more detailed description herein.

(75) A problem with (conventional) offset calibration as discussed in connection with FIG. 4 lies in that calibration is performed at a specific/instantaneous value of the power supply voltage V_M which can then vary significantly during normal system operation.

(76) It is observed that the power supply voltage of an HDD can undergo variations of $\pm 10\%$ or higher, consequently producing variations in the common-mode voltage present across the sense amplifier 16 (equal or very close to $V_M/2$ during normal VCM operations).

(77) The sense amplifier is in fact characterized by a common-mode rejection ratio that has a finite (non-infinite) value. This results in a voltage at the output of the sense amplifier 16 that is a function of the common-mode input voltage.

(78) A conventional calibration approach as discussed in connection with FIG. 4 does not take into account possible offset variations caused by the variation of the common mode voltage present at the input of the sense amplifier 16.

(79) Once such a (conventional) VCM offset calibration (offset measurement) is performed, the resulting value is kept fixed during subsequent offset compensation (as represented in dashed lines in FIG. 4) independently of any variations in the power supply level V_M .

(80) Changes in the power supply level result in changes in the common mode at the inputs of the sense amplifier 16, which in turn result in variations of the output of the sense amplifier 16 that will be not adequately compensated by the offset calibrated (measured) value. The VCM current will be shifted with respect to the desired value computed by the VCM position control loop (microcontroller) and imposed through the DAC input word DW.

(81) This offset variation is a source of nuisance that produces unwanted effects such as an increased run-out (the error during track following).

(82) A solution as discussed herein in connection with FIGS. 5A, 5B (offset measurement/calibration) and 6 (offset compensation), plus FIGS. 7 to 9 is able to reduce the error present on the zero VCM current offset calibration value (performed at fixed power supply level) irrespective of possible changes in the power supply level in an operating range of, e.g., to about 10V to 14V).

(83) Various solutions as discussed herein in connection with FIGS. 5A, 5B, and 6 to 9 facilitate taking into account offset variations in a transconductance amplifier that occur as a function of the value of the power supply voltage. Accuracy in track following is thus improved also in the presence of a power supply voltage that is not stable.

(84) In the following, parts or elements like parts or elements already discussed in connection with FIGS. 1 through 4 are indicated with like reference symbols: a detailed description of these parts or elements as already provided also applies to the subsequent figures and will not be repeated for brevity.

(85) Solutions as discussed in connection with 5A, 5B, and 6 to 9 address the problem of offset variation that may occur due to the variation of the supply voltage V_M of a transconductance amplifier.

(86) As noted, the common-mode voltage at the drive stage output (and therefore the voltage at the inputs of the sense amplifier 16) is usually forced to $V_M/2$ (half the supply voltage V_M) in order to exploit the full dynamics of the power stage. This however implies a corresponding variation as the supply voltage V_M varies. The variation of the supply voltage V_M and therefore of $V_M/2$ produces undesired variations on the output voltage of the sense amplifier 16 and therefore an offset variation in the transconductance amplifier.

(87) It is noted that the variation of the transconductance amplifier offset is due to the variation of the output of the sense amplifier 16 due to its limited CMRR. A change in common-mode voltage at the inputs of the sense amplifier 16 results in a (small, yet not nil) change in the voltage at the output of the sense amplifier 16, and this results in a non-constant offset in the current controlled by the transconductance loop.

(88) It is observed that the common mode voltage value ($V_m/2$) is a function of the supply voltage VM, which facilitates compensating with a feed-forward approach the variations of the offset of the transconductance loop using the direct measurement of the supply voltage and correcting dynamically the value of the offset to insert in the loop.

(89) FIGS. 5A and 5B show an example of calibration (essentially, offset measurement) according to embodiments of the present description, where two different common-mode voltages are forced to the inputs of the sense amplifier **16** in order to evaluate their effects (in terms of offset) on the transconductance loop.

(90) Embodiments of the present description can thus be applied to operating (at least) one voice coil motor VCM in a hard disk drive via a transconductance loop **10**, **12**, **16** having a supply voltage VM.

(91) It is once again reminded that the representation in FIGS. 4, 5A, 5B and 6 is essentially functional, in that the motor VCM and the parts/elements (primarily the transconductance loop) associated therewith can in fact be incorporated in the hard disk drive HDD, while the associated controller MCU that controls operation of the HDD as discussed herein can be either incorporated in the hard disk drive HDD or be a distinct element.

(92) The transconductance loop as illustrated in figures such as FIGS. 5A, 5B and 6 again comprises: a power stage **10** configured (see, e.g., the stages **101** and **102** producing the signals VCMN and VCMP) to apply a drive current to the voice coil motor VCM, a differential sense amplifier **16** having first and second inputs coupled to a current sensor (e.g., a resistor R_s) of the voice coil motor VCM wherein the sense amplifier **16** is configured to produce a sense signal indicative of the drive current $I_{sub.vcm}$ applied to the voice coil motor VCM.

(93) Likewise, the transconductance loop as illustrated in FIGS. 5A, 5B and 6 comprises an error amplifier **12** configured to receive a loop control signal (such as a digital word DW received via a digital-to-analog, DAC converter **14**, and possibly compensated as discussed in the following) to produce a “target” (namely, a desired) drive current for the voice coil motor VCM.

(94) As illustrated, the error amplifier **12** is coupled to the power stage **10** to drive the power stage based on the loop control signal (essentially the digital word DW as received once converted to analog by the DAC converter **14**) and the sense signal.

(95) As noted, such a transconductance loop and its basic principle of operation are per se conventional, which makes it unnecessary to provide a more detailed description herein.

(96) Embodiments of the present description can be implemented using virtually any number of (at least two) measurements and (at least two) values of the voltages across the inputs of the sense amplifier **16** used for the implementation.

(97) For simplicity and ease of understanding, FIGS. 5A and 5B refer to an example of calibration where two offset measurements with two different voltage values are performed.

(98) This facilitates establishing, based a first offset value and a second offset value, a relationship between offset values and supply voltages of the transconductance loop, namely calculating a characteristic function which, in a simple case, is the equation of a straight line passing through two (measured) points.

(99) Based thereon (as further discussed in the following) an offset compensation signal **20A** (see FIG. 6) of the transconductance loop can be calculated to be applied at summation node **18** at the input of the DAC converter **14** to the loop control signal DW of the transconductance loop **10**, **12**, **16**.

(100) Such offset compensation (injection of the signal OC into the input word DW at the node **18**) takes place with the transconductance loop **10**, **12**, **16** brought to its normal operating condition.

(101) FIGS. 5A and 5B thus refer to an example of calibration according to embodiments of the present description where two offset measurements are used, obtained with two different common-mode voltages forced at the sense amplifier **16** inputs.

(102) FIG. 5A shows an example of a circuit suitable for forcing a common-mode voltage equal to

$V_m/2$, that is, equal to half V_M , while FIG. 5B shows an example of a circuit suitable for forcing a common-mode voltage V_m equal to V_M .

(103) This type of operation is again facilitated (as in the case of FIG. 4) via the switch **Sw2** that, when non-conductive, decouples the stage **102** from the node A, thus from the error amplifier **12**. It is noted that the other stage **101** is forced to high impedance to have zero current in the load.

(104) This type of operation is likewise facilitated by a modified arrangement of the switch **Sw3** and the addition of a further switch **Sw4** (these can again be implemented in a manner known per se to those of skill in the art, e.g., as a MOSFET transistor) configured to: couple the non-inverting input of the stage **102** to the inverting input of the stage **102** (with the switch **Sw3** closed in a first position, e.g., **Sw3-A**, and **Sw4** closed) so that a voltage $V_M/2$ results at the stage **102** output—see FIG. 5A; decouple the non-inverting input of the stage **102** from the inverting input (with the switch **Sw3** closed in a second position, e.g., **Sw3-B**, and **Sw4** closed) so that a voltage V_M results at the stage **102** output—see FIG. 5B.

(105) Here again, a switch is regarded as closed when brought to conductive state.

(106) It is otherwise noted that the values referred herein ($V_m/2=V_M/2$ and $V_m=V_M$, respectively) are merely indicative of a pair of values that can be used to identify two different transconductance loop offset values. Other (pairs of) values can be used: $V_m=V_M$ and GND, respectively, are another possible example.

(107) Again, during calibration of the offset of the transconductance amplifier (offset measurement) as exemplified in FIGS. 5A and 5B, the gain of the error amplifier **12** can be (markedly) increased by opening the switch **Sw1** in the feedback path of the error amplifier **12**.

(108) This increased gain facilitates the detection of changes of sign of the error signal Error OUT which is carried out by analyzing the output of the error amplifier available over the line Error OUT.

(109) As noted, under ideal conditions (zero offset) the signal on the line Error OUT changes polarity when the input word DW to the DAC converter **14** moves around zero and, as consequence, the output signal from the DAC converter **14** moves around the reference value V_{ref} (again, as illustrated in FIGS. 5A, 5B and 6, the DAC converter **14** is configured to have an output voltage equal to V_{ref} when its input word DW is zero).

(110) In real conditions (see again the diagram of FIG. 3 for reference) the change of polarity (sign) of the error signal Error OUT takes place instead for values of the input word DW other than zero (for DAC output other than V_{ref}), which depend on the offset of the sense amplifier **16** and the offset of the error amplifier **12**.

(111) A calibration procedure as exemplified in FIGS. 5A and 5B again involves a search for values (e.g., a first and a second value) for the input word DW that, if applied to the DAC converter **14**, produce a change of sign (polarity) of the output of the error amplifier (line Error OUT).

(112) Once these values for the offset voltage and the supply voltage have been measured (the latter can be made available thanks to a dedicated ADC converter that is usually available in HDD power combos) the possibility will then exist of implementing (e.g., via the controller MCU, based on the signal on the line Error OUT) the function which links the offset value of the transconductance loop to the value V_m .

(113) If for example, as in FIGS. 5A and 5B, the two common-mode voltages used for the double offset measurement are $V_m/2=V_M/2$ and $V_m=V_M$ it is possible, based on the value for V_M using the simple calculation of the function of a straight line passing from two points, calculate the value of the offset to insert in the circuit for each value of V_M in order to restore a situation like the one shown in FIG. 2 and valid for any value of V_M voltage.

(114) To summarize, a calibration as discussed herein comprises (at least) one first calibration condition—as exemplified in FIG. 5A—and (at least) one second calibration condition—as exemplified in FIG. 5B.

(115) In the first calibration condition, the first and second inputs of the sense amplifier **16** have

applied thereto a first common input voltage (e.g., $V_m/2=V_m/2$) and a first value of the DAC input word DW is determined which results—in such a first calibration condition—in a change of sign (polarity) of the error signal Error OUT.

(116) In the second calibration condition, the first and second inputs of the sense amplifier **16** have applied thereto a second common input voltage VM and a second value of the DAC input word DW is determined which results—in such a second calibration condition—in a change of sign (polarity) of the error signal Error OUT.

(117) The second common input voltage VM is different from the first common input voltage VM/2 and at least one (and optionally both) of the first and second common input voltages is/are a function of the supply voltage VM of the transconductance **10, 12, 16**).

(118) Based on the first value and the second value of the DAC input word DW that were found to result—in the first and the second calibration condition, respectively—in a change of sign (polarity) of the error signal Error OUT, a relationship is established between these two “offset” values (as expressed, for instance, in terms of values of DAC input word DW) and the supply voltage VM: this is feasible relying on the fact that at least one (and optionally both) of the first and second common input voltages is/are a function of the supply voltage VM of the transconductance loop.

(119) An actual value of the supply voltage VM can thus be detected and an offset compensation signal **20A** of the transconductance loop can be produced as a function of the actual detected value of the supply voltage VM based the relationship (e.g., a straight line) between the (at least) two values for the DAC input word DW measured during offset calibration and the supply voltage VM.

(120) As exemplified in FIG. 6, an offset compensation (cancellation) signal **20A** of the transconductance loop **10, 12, 16** can thus be applied (at a node **18**, for instance) to the loop control signal DW, namely the DAC input word DW indicative of the (desired) target drive current for the voice coil motor VCM (e.g., in order to achieve a zero current $I_{sub.VCM}$ for a zero input word DW).

(121) To summarize, various embodiments as discussed herein involve at least one first calibration condition and at least one second calibration condition.

(122) In the first calibration condition (as exemplified in FIG. 5A), the first and second inputs of the sense amplifier **16** have applied thereto a first common input voltage (VM/2) and the error amplifier **12** has applied thereto the loop control signal (the digital word DW received via the DAC converter **14**, for instance) with a first (offset) value that produces a commutation in the sign of the error signal Error OUT at the output of the error amplifier **12** in the first calibration condition.

(123) In the second calibration condition (as exemplified in FIG. 5B) the first and second inputs of the sense amplifier **16** have applied thereto a second common input voltage VM and the error amplifier **12** has applied thereto the loop control signal DW with a second (offset) value that produces a commutation in the sign of the error signal Error OUT at the output of the error amplifier **12** in the second calibration condition.

(124) The second common input voltage VM is different from the first common input voltage VM/2 and at least one (and advantageously both) of the first and second common input voltages is/are a function of the supply voltage VM of the transconductance loop **10, 12, 16**.

(125) Again, in an arrangement as exemplified in FIGS. 5A, 5B, and 6, the error signal Error OUT (that drives via the node A the power stage **10** and thus the load represented by the motor VCM) depends on the difference between the two inputs of the error amplifier **12**, e.g.: a non-inverting input, coupled (e.g., via a resistor) to a reference node at voltage Vref; and an inverting input, coupled (e.g., via the resistor Rin) to the DAC converter **14** as well as (e.g., via the resistor Rf) to the output of the sense amplifier.

(126) Based on the first offset value and the second offset value, a relationship is established between offset values and the supply voltage VM of the transconductance loop **10, 12, 16**.

(127) An actual value of the supply voltage VM of the transconductance loop **10, 12, 16** is detected

and an offset compensation signal **20A** of the transconductance loop **10**, **12**, **16** is produced as a function of the actual detected value of the supply voltage VM based on the relationship established between the offset values and supply voltage VM of the transconductance loop **10**, **12**, **16**.

(128) Such an offset compensation signal can thus be applied (e.g., at the summation node **18**—with sign—at the input of the DAC converter **14** as exemplified in FIG. **6**) to the loop control signal DW of the transconductance loop **10**, **12**, **16**.

(129) As illustrated (during the calibration phases exemplified in FIGS. **5A** and **5B**), the power stage **10** can be at least partially de-activated (stage **101** in HiZ state) to discontinue supplying drive current to the voice coil motor VCM during the first calibration condition and the second calibration condition.

(130) FIG. **6** shows a possible practical implementation of how the results of the calibration phases exemplified in FIGS. **5A** and **5B** can be exploited for offset compensation of the transconductance loop.

(131) FIG. **6** shows the switches involved in measuring the (at least) two offset values during the calibration procedure (namely, the switches Sw**1**, Sw**2**, Sw**3**, Sw**4** to be operated as described previously in connection with FIGS. **5A** and **5B**) and the circuits (namely the elements indicated by the reference numerals **18**, **20**, **22A**, **22B**, **22C**) used to dynamically compensate the offset during the normal VCM driving activity—e.g., tracking—as represented in FIG. **6**.

(132) Again, the switches Sw**1**, Sw**2**, Sw**3**, Sw**4** can be implemented in a manner known per se to those of skill in the art, e.g., as a MOSFET transistors and can be operated (turned on=conductive and off=non-conductive) under the control of the controller MCU, for instance.

(133) It is noted that in FIG. **6** the switches Sw**1**, Sw**2**, Sw**3**, Sw**4** are illustrated in the condition used for normal VCM activity (that is after calibration/calculation of the offset v. supply voltage characteristic curve).

(134) As illustrated in FIG. **6**, the input value to the DAC converter **14** is provided by the summation node **18**, namely as the input word DW “compensated” by the dynamic offset compensation value ([X:0]) **20A** that is obtained in an offset computation block **20**.

(135) The block **20** receives as inputs the two offset values ([X:0]) calculated during calibration with the two common-mode voltages used for the double offset measurement, e.g., $V_m/2=VM/2$ and $V_m=VM$.

(136) In various embodiments the block **20** may not be configured to calculate these values per se, but rather to receive these two offset values that are calculated as discussed previously during the offset calibration phase: this can be done, e.g., via the (possibly external) microcontroller MCU usually associated with the hard disk drive HDD (see also the flow chart presented in FIG. **8**).

(137) The block **20** processes these two values to obtain, e.g., in real-time, an offset value to be applied to the loop during normal operation of the VCM.

(138) As discussed, offset calculation can take place—e.g., in the controller MCU, in a manner known per se to those of skill in the art—based on the same criteria used to calculate a (single) offset value in the conventional calibration procedure discussed in connection with FIG. **4**.

(139) In the exemplary solution presented herein that calculation takes place twice, namely with the common-mode voltages at the input of the sense amplifier set at, e.g., $V_m/2=VM/2$ and $V_m=VM$.

(140) The block **20** uses the two offset values received to find a correct offset **20A** to be summed (at the node **18**) to the DAC input word DW in order to produce an I.sub.VCM v. DW characteristic as close as possible to that shown in FIG. **2** independently by the VM voltage.

(141) In that way a dynamic offset characteristic as portrayed in FIG. **7** can be obtained—starting from two measurement points indicated as I ($V_m/2=VM/2$) and II ($V_m=VM$): FIG. **7** that portrays a possible behavior of a loop offset compensation value (e.g., in terms of DAC word DW—decimal) versus the value for VM.

(142) The value for VM measured through a “VM ADC” converter **22C** (e.g., [X:0]) can be

supplied to the DAC offset computation block **20**. The DAC offset computation block **20** will thus provide an adequate offset compensation value according to the “instantaneous” value of VM even if this value varies, e.g., within an HDD operating voltage range, e.g., between about 10 and 14 Volt.

(143) A desired dynamic offset compensation resolution can be achieved according to an application specification (e.g., in terms of VM ADC bit resolution).

(144) The flow chart in FIG. **8** is exemplary of a possible implementation of:

(145) An offset calibration (measurement) procedure as discussed in the foregoing (left-hand side of FIG. **8**), and

(146) An offset compensation procedure during normal operation of the HDD power combo (right-hand side of FIG. **8**).

(147) Switching between the two involves controlling the switches Sw**1**, Sw**2**, Sw**3**, and Sw**4**.

(148) An offset calibration and compensation procedures as discussed herein can be activated in a power combo as repeatedly discussed herein in order to reduce (notionally cancel) the VCM current control loop offset.

(149) That calibration routine can be performed, e.g., in response to a command from the HDD controller, for instance before tracking operation of the heads is started or whenever held advantageous under the control of the controller MCU.

(150) After a START (command/event) as illustrated on the left-hand side of FIG. **8** in steps as represented by blocks **1000** and **1002** in FIG. **8**, the transconductance loop is operated first in the condition represented in FIG. **5A** (e.g., $V_m/2=VM/2$) and then in the condition represented in FIG. **5B** (e.g., $V_m=VM$) to produce (at least) two offset measurement points (e.g., I and II in FIG. **7**).

(151) In a step as represented by block **1004** in FIG. **8** a (linear) characteristic as illustrated in FIG. **7** can be determined (calculated) starting from these points. Such a characteristic lends itself to being stored, e.g., as a look-up table to the DAC offset computation block **20** or to being calculated based on a function that links the common mode voltage to a related offset.

(152) Once such an offset calibration procedure completed, in a step as represented by block **1006** on the right-hand side of FIG. **8**, a current value for VM (that is V_m) is measured (e.g., through the VM ADC converter **22C**) and supplied to the DAC offset computation block **20**.

(153) In a step as represented by block **1008** in FIG. **8**, the DAC offset computation block **20** is thus in a position to provide an adequate offset compensation value according to the instantaneous value of VM even if this value varies within an HDD operating voltage range.

(154) FIG. **8** is not otherwise to be construed as implying by way of necessity a one-to-one relationship between offset calibration procedure and normal operation of the HDD: in fact, offset compensation can be activated with a high refresh rate when controlling a VCM.

(155) FIGS. **9A** and **9B** show the effects, in terms of current in the load (assuming a constant value for the DAC input word DW is applied) of possible variations of the voltage VM: in the case of a conventional arrangement (FIG. **4** and FIG. **9A**), and in the case of an arrangement using dynamic compensation as discussed herein (FIGS. **5A**, **5B**, **6**, **7**, and **8** and FIG. **9B**).

(156) The curves in FIGS. **9A** and **9B** are exemplary of possible time behaviors of the following signals (from top to bottom): the supply voltage VM, the drive voltage on line VCMP, the drive voltage on line VCMN, the current $I_{sub.CM}$.

(157) The lowest curves show the effects, in terms of current in the load, of the variation of the voltage V_m in the case of a conventional arrangement (FIG. **9A**), and in the case of an arrangement using dynamic compensation as discussed herein (FIG. **9B**).

(158) In these figures the desired current value is constant (the DAC Word DW is assumed to be constant).

(159) Due to variation of offset, the current in the load is not constant in the case of a conventional arrangement as portrayed in FIG. **9A**.

(160) Conversely, in an arrangement using dynamic compensation as discussed herein the current is

substantially constant as portrayed in FIG. 9B.

(161) Of course, the time behaviors indicated/shown in FIGS. 9A and 9B are purely indicative and non-limiting.

(162) Without prejudice to the underlying principles, the details and embodiments may vary, even significantly, with respect to what has been described in the foregoing, by way of example only, without departing from the extent of protection.

(163) The extent of protection is determined by the annexed claims.

Claims

1. A method of operating a voice coil motor via a transconductance loop, the method comprising: detecting an actual value of a supply voltage of the transconductance loop; producing an offset compensation signal of the transconductance loop as a function of the detected actual value of the supply voltage based on a relationship between offset values and the supply voltage of the transconductance loop; applying the offset compensation signal to a loop control signal of the transconductance loop, wherein a target drive current for the voice coil motor is based on the loop control signal; and applying a drive current to the voice coil motor, the drive current related to the target drive current.
2. The method of claim 1, wherein the offset values comprise a first offset value that produces a commutation in a sign of an error signal at an output of an error amplifier of the transconductance loop in a first calibration condition, and a second offset value that produces a commutation in the sign of the error signal at the output of the error amplifier in a second calibration condition.
3. The method of claim 2, wherein the drive current is applied by a power stage of the transconductance loop, the method further comprising at least partially de-activating the power stage to discontinue supplying drive current to the voice coil motor during the first calibration condition and the second calibration condition.
4. The method of claim 2, wherein the relationship between the offset values and the supply voltage of the transconductance loop is a straight line passing through the first offset value and the second offset value.
5. The method of claim 2, wherein producing the offset compensation signal of the transconductance loop comprises reading the offset compensation signal in a look-up table having stored therein the relationship between the offset values and the supply voltage of the transconductance loop and having entries for the detected actual value of the supply voltage.
6. The method of claim 2, wherein producing the offset compensation signal of the transconductance loop comprises calculating a relationship between the first offset value and the second offset value and first and second common input voltages applied to first and second inputs of a sense amplifier of the transconductance loop.
7. A transconductance loop circuit configured to operate with the method of claim 1, the transconductance loop comprising: a power stage configured to apply a drive current to the voice coil motor; a differential sense amplifier having first and second inputs configured to be coupled to a current sensor of the voice coil motor, wherein the differential sense amplifier is configured to produce a sense signal indicative of the drive current applied to the voice coil motor; and an error amplifier configured to receive the loop control signal to produce the target drive current for the voice coil motor, wherein the error amplifier is coupled to the power stage to drive the power stage based on the loop control signal and the sense signal.
8. A method of operating a voice coil motor in a hard disk drive via a transconductance loop having a supply voltage, wherein the transconductance loop comprises a power stage configured to apply a drive current to the voice coil motor, a differential sense amplifier having first and second inputs coupled to a current sensor of the voice coil motor, wherein the differential sense amplifier is configured to produce a sense signal indicative of the drive current applied to the voice coil motor,

and the transconductance loop comprises an error amplifier configured to receive a loop control signal to produce a target drive current for the voice coil motor, wherein the error amplifier is coupled to the power stage to drive the power stage based on the loop control signal and the sense signal, wherein the method comprises: performing a calibration with a first calibration condition and a second calibration condition of the transconductance loop wherein, in the first calibration condition, the first and second inputs of the differential sense amplifier have applied thereto a first common input voltage and the error amplifier has applied thereto said loop control signal with a first offset value that produces a commutation in a sign of an error signal at an output of the error amplifier in the first calibration condition, and, in the second calibration condition, the first and second inputs of the differential sense amplifier have applied thereto a second common input voltage and the error amplifier has applied thereto said loop control signal with a second offset value that produces a commutation in the sign of the error signal at the output of the error amplifier in the second calibration condition, wherein the second common input voltage is different from the first common input voltage and at least one of the first and second common input voltages is a function of the supply voltage of the transconductance loop; establishing, based on the first offset value and the second offset value, a relationship between offset values and supply voltage of the transconductance loop; detecting an actual value of the supply voltage of the transconductance loop; producing an offset compensation signal of the transconductance loop as a function of the detected actual value of the supply voltage based on the relationship between the offset values and the supply voltage of the transconductance loop; and applying to the loop control signal of the transconductance loop the offset compensation signal.

9. The method of claim 8, wherein both of the first and second common input voltages applied to the first and second inputs of the differential sense amplifier are a function of the supply voltage of the transconductance loop.

10. The method of claim 8, comprising at least partially de-activating the power stage to discontinue supplying drive current to the voice coil motor during the first calibration condition and the second calibration condition.

11. The method of claim 8, wherein the relationship between offset values and the supply voltage of the transconductance loop is a straight line passing through the first offset value and the second offset value.

12. The method of claim 8, further comprising producing the offset compensation signal of the transconductance loop by reading it in a look-up table having stored therein said relationship between offset values and the supply voltage of the transconductance loop and having entries for the detected actual value of the supply voltage.

13. The method of claim 8, further comprising producing the offset compensation signal of the transconductance loop by calculating a relationship between the first offset value and the second offset value and the first and second common input voltages applied to the first and second inputs of the differential sense amplifier.

14. A transconductance loop circuit comprising: a power stage configured to apply a drive current to a voice coil motor in a hard disk drive; a differential sense amplifier having first and second inputs configured to be coupled to a current sensor of the voice coil motor, wherein the differential sense amplifier is configured to produce a sense signal indicative of the drive current applied to the voice coil motor; an error amplifier configured to receive a loop control signal to produce a target drive current for the voice coil motor, wherein the error amplifier is coupled to the power stage to drive the power stage based on the loop control signal and the sense signal; offset calibration circuitry configured to be brought to a first calibration condition and a second calibration condition and to establish a relationship between offset values and a supply voltage of the transconductance loop circuit, based on a first offset value and a second offset value; a detector configured to detect an actual value of the supply voltage; a processing circuit configured to produce an offset compensation signal of the transconductance loop circuit as a function of the detected actual value

of the supply voltage based on the relationship between the offset values and the supply voltage of the transconductance loop circuit; and an offset compensation node configured to apply the offset compensation signal to the loop control signal of the transconductance loop circuit.

15. The transconductance loop circuit of claim 14, wherein the power stage is configured to be at least partially de-activated to discontinue supplying drive current to the voice coil motor during the first calibration condition and the second calibration condition.

16. The transconductance loop circuit of claim 14, wherein the processing circuit is configured to produce the offset compensation signal of the transconductance loop circuit by reading it in a look-up table having stored therein said relationship between the offset values and the supply voltage of the transconductance loop circuit and having entries for the detected actual value of the supply voltage.

17. The transconductance loop circuit of claim 14, wherein the processing circuit is configured to produce the offset compensation signal of the transconductance loop circuit by calculating a relationship between the first offset value and the second offset value and first and second common input voltages applied to the first and second inputs of the differential sense amplifier.

18. A power combo integrated circuit for use with a hard disk drive that includes a hard disk, a spindle motor, and a voice coil motor, the power combo integrated circuit comprising: the transconductance loop circuit according to claim 14; and a spindle motor driver; wherein the power combo integrated circuit is configured to operate with the hard disk drive such that the spindle motor driver drives the spindle motor to cause the hard disk to spin and the transconductance loop circuit drives the voice coil motor to position a read/write head at a surface of the hard disk.

19. A hard disk drive, comprising: a hard disk; a voice coil motor functionally coupled to the hard disk; and the transconductance loop circuit according to claim 14, wherein the power stage is coupled to the voice coil motor to apply the drive current to the voice coil motor.

20. The hard disk drive of claim 19, further comprising: a spindle motor functionally coupled to the hard disk to cause the hard disk to spin, wherein the voice coil motor is functionally coupled to the hard disk to position a read/write head at a surface of the hard disk; and a spindle motor driver coupled to the spindle motor to apply a spindle motor drive current to the spindle motor.
